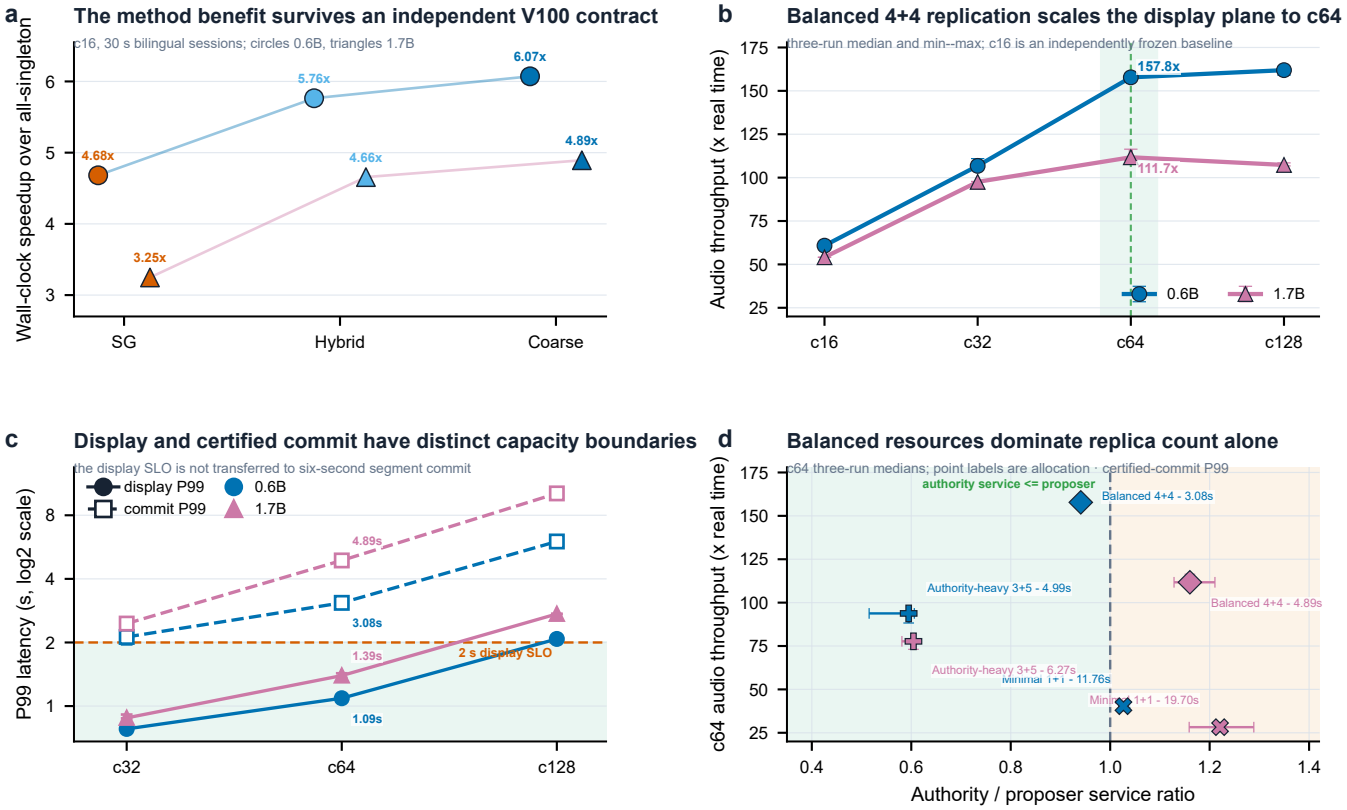

a
The method benefit survives an independent V100 contract
c16, 30 s bilingual sessions; circles 0.6B, triangles 1.7B
Wall-clock speedup over all-singleton
6.07x
5.76x
4.89x
4.66x
4.68x
3.25x
SG
Hybrid
Coarse
b
Balanced 4+4 replication scales the display plane to c64
three-run median and min--max; c16 is an independently frozen baseline
Audio throughput (x real time)
157.8x
111.7x
0.6B
1.7B
c16
c32
c64
c128
c
Display and certified commit have distinct capacity boundaries
the display SLO is not transferred to six-second segment commit
display P99
commit P99
0.6B
1.7B
P99 latency (s, log2 scale)
4.89s
3.08s
1.39s
1.09s
2 s display SLO
c32
c64
c128
d
Balanced resources dominate replica count alone
c64 three-run medians; point labels are allocation · certified-commit P99
c64 audio throughput (x real time)
authority service <= proposer
Balanced 4+4 - 3.08s
Authority-heavy 3+5 - 4.99s
Balanced 4+4 - 4.89s
Authority-heavy 3+5 - 6.27s
Minimal 1+1 - 11.76s
Minimal 1+1 - 19.70s
Authority / proposer service ratio